Software Qualification – RC Laser NA265280

Contents

Notes:

- **All the measurements are made with respect to GND.**
- **Test point numbers and labels are for PCBA Rev F.**

1. Purpose

To provide a test procedure for verifying that all software functions associated with the RC Laser NA265280 are fully functional as designed. This qualification is specific to the RC Laser.

2. Explanation of Terminology

The RC Laser NA265280 is a remote control laser system designed to save time and effort when adjusting the placement of a laser line or lines. It has three line lasers (plumb 1, plumb 2, and level), which can be adjusted vertically and horizontally. The unit uses Bluetooth Low Energy technology for remote operation from a distance of up to 100 meters (about 330 feet) between the laser unit and remote control. The user can also adjust laser brightness.

3. Equipment required.

Oscilloscope, RC Laser NA265280 Module programmed, Source Meter, Stopwatch, 20V DeWalt Battery

4. Laser

4.1 Laser Level

Item Tested:	Level Laser PWM frequency
Item Spec:	Level laser Turns ON with a Pulse Train for Detector Frequency 1. Laser Turn ON 500ms at 200Khz at 100% dutycycle 2. After 500ms, Laser is ON at 8.5Khz at 50% duty for 4ms
Equipment needed:	RC Laser module flashed and programmed, Oscilloscope, Dewalt Battery, Power Supply.
Pass / Fail Criteria	**Pass:** 1. Laser turn on at desired PWM and outputs a perfect pulse train for detector. **Fail:** 1. Laser doesn't Turn ON 2. The PWM duty cycle is out of spec.

<table>
<tr><td>Test Procedure:</td><td>

Test Procedure:

1. Connect a wire to Test point I129 (net label L). Connect this to oscilloscope to measure freq.
2. Set up the oscilloscope to monitor the PWM frequency and duty cycle of the pulse train (200Khz for 500ms and then 8.5Khz for 4ms).
3. Connect RC Laser Battery terminal to a Power supply.
4. Connect a 10kOhms resistor between B+ and TH on RC Laser Battery terminal.
5. Set the power supply to output 20V@300mA

</td></tr>
<tr><td>Test Procedure:</td><td>

1. Set the Power Supply voltage to 20V@300mA.
2. Slide the pendulum lock to unlocked position.
3. Press the power Button.
4. Press the laser Level Button if the level laser doesn’t turn ON by default.
5. Measure and record the data in the result table below.
6. Repeat the process down to 15V with a decrement interval of 1V.

Output:

S.no	Voltage	State of Laser	Main Freq and time	Detector freq and time
1	20			
2	19			
3	18			
4	17			
5	16			

</td></tr>
</table>

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

4.2 Laser Plumb1

Item Tested:	Level Plumb1 PWM frequency
Item Spec:	Level laser Turns ON with a Pulse Train for Detector Frequency 1. Laser Turn ON 500ms at 200Khz at 100% dutycycle 2. After 500ms, Laser is ON at 8.5Khz at 50% duty for 4ms
Equipment needed:	RC Laser module flashed and programmed, Oscilloscope, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. Laser turn on at desired PWM and outputs a perfect pulse train for detector. **Fail:** 1. Laser doesn't Turn ON 2. The PWM duty cycle is out of spec.
Test Procedure:	**Test Procedure:** 1. Connect a wire to Test point I125 (net label P2/D1). Connect this to oscilloscope to measure freq. 2. Set up the oscilloscope to monitor the PWM frequency and duty cycle of the pulse train (200Khz for 500ms and then 8.5Khz for 4ms). 3. Connect RC Laser Battery terminal to a Power supply. 4. Connect a 10kOhms resistor between B+ and TH on RC Laser Battery terminal. 5. Set the power supply to output 20V@300mA

Test Procedure:	1. Set the voltage to 20V@300mA. 2. Slide the pendulum lock to unlocked position. 3. Press the power Button on the Laser Unit 4. Turn off any lasers that are on. 5. Press the laser Plumb1 Button. 6. Measure and record the data in result table below. 7. Repeat the process down to 15V with decrement of 1V. **Output:**			
Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

Output table:

S.no	Voltage	State of Laser	Main Freq and time	Detector freq and time
1	20			
2	19			
3	18			
4	17			
5	16			

4.3 Laser Plumb2

Item Tested:	Level Plumb2 PWM Frequency
Item Spec:	Level laser Turns ON with a Pulse Train for Detector Frequency 1. Laser Turn ON 500ms at 200Khz at 100% dutycycle 2. After 500ms, Laser is ON at 8.5Khz at 50% duty for 4ms
Equipment needed:	RC Laser module flashed and programmed, Oscilloscope, Dewalt Battery

Pass / Fail Criteria	**Pass:** 1. Laser turn on at desired PWM and outputs a perfect pulse train for detector. **Fail:** 1. Laser doesn't Turn ON 2. The PWM duty cycle is out of spec.
Test Procedure:	**Test Procedure:** 1. Connect a wire to Test point I133 (net label P1). Connect this to oscilloscope to measure freq. 2. Set up the oscilloscope to monitor the PWM frequency and duty cycle of the pulse train (200Khz for 500ms and then 8.5Khz for 4ms). 3. Connect RC Laser Battery terminal to a Power supply. 4. Connect a 10kOhms resistor between B+ and TH on RC Laser Battery terminal. 5. Set the power supply to output 20V@300mA
Test Procedure:	1. Set the voltage to 20V@300mA. 2. Slide the pendulum lock to unlocked position. 3. Press the power Button on the Laser Unit 4. Turn off any lasers that are on. 5. Press the laser Plumb2 Button. 6. Measure and record the data in result table below. 7. Repeat the process down to 15V with decrement of 1V.

Output:

S.no	Voltage	State of Laser	Main Freq and time	Detector freq and time
1	20			
2	19			
3	18			
4	17			
5	16			

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

4.4 Laser Level Brightness

Item Tested:	Laser Level Brightness PWM Duty cycle
Item Spec:	Laser level cycles duty cycle between 100, 50, 35 when the button is pressed.
Equipment needed:	RC Laser module flashed and programmed, Oscilloscope, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. Laser turns on at desired PWM and outputs a perfect pulse train for detector. **Fail:** 1. Laser doesn't Turn ON 2. The PWM duty cycle is out of spec.
Test Procedure:	**Test Procedure:** 1. Connect a wire to Test point I129 (net label L). Connect this to oscilloscope to measure freq. 2. Set up the oscilloscope to monitor the PWM frequency and duty cycle of the pulse train (200Khz for 500ms and then 8.5Khz for 4ms). 3. Connect RC Laser Battery terminal to a Power supply. 4. Connect a 10kOhms resistor between B+ and TH on RC Laser Battery terminal. 5. Set the power supply to output 20V@300mA

Test Procedure:	1. Insert a charged 20V DeWalt battery pack into the laser unit. 2. Slide the pendulum lock to unlocked position. 3. Press the power button. 4. Press the laser Level Button. 5. Press the brightness Button. 6. Measure and record the data in result table below. 7. Repeat the process 5 times for each brightness level. i.e Press the Brightness button 15 times. **Output:**

S.no	Duty Cycle	Case
1		Test 1
2		
3		
4		Test 2
5		
6		
7		Test 3
8		
9		
10		Test 4
11		
12		
13		Test 5
14		
15		

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

4.5 Laser Plumb1 Brightness

Item Tested:	Laser Plum1 Brightness PWM Duty cycle
Item Spec:	Laser level cycles duty cycle between 100, 75, 50 when the button is pressed.
Equipment needed:	RC Laser module flashed and programmed, Oscilloscope, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. Laser turns on at desired PWM and outputs a perfect pulse train for detector. **Fail:** 1. Laser doesn't Turn ON 2. The PWM duty cycle is out of spec.
Test Procedure:	**Test Procedure:** 1. Connect a wire to Test point I125 (net label P2/D1). Connect this to oscilloscope to measure freq. 2. Set up the oscilloscope to monitor the PWM frequency and duty cycle of the pulse train (200Khz for 500ms and then 8.5Khz for 4ms). 3. Connect RC Laser Battery terminal to a Power supply. 4. Connect a 10kOhms resistor between B+ and TH on RC Laser Battery terminal. 5. Set the power supply to output 20V@300mA
Test Procedure:	1. Insert a charged 20V DeWalt battery pack into the laser unit. 2. Slide the pendulum lock to unlocked position. 3. Press the power button.

<table>
<tr><td></td><td>
4. Turn off any laser that are On.
5. Press the laser Plumb1 Button.
6. Press the brightness Button.
7. Measure and record the data in result table below.
8. Repeat the process 5 time for each brightness level. i.e Press the Brightness button 15 times.

Output:
<table>
<tr><th>S.no</th><th>Duty Cycle</th><th>Case</th></tr>
<tr><td>1</td><td></td><td rowspan="3">Test 1</td></tr>
<tr><td>2</td><td></td></tr>
<tr><td>3</td><td></td></tr>
<tr><td>4</td><td></td><td rowspan="3">Test 2</td></tr>
<tr><td>5</td><td></td></tr>
<tr><td>6</td><td></td></tr>
<tr><td>7</td><td></td><td rowspan="3">Test 3</td></tr>
<tr><td>8</td><td></td></tr>
<tr><td>9</td><td></td></tr>
<tr><td>10</td><td></td><td rowspan="3">Test 4</td></tr>
<tr><td>11</td><td></td></tr>
<tr><td>12</td><td></td></tr>
<tr><td>13</td><td></td><td rowspan="3">Test 5</td></tr>
<tr><td>14</td><td></td></tr>
<tr><td>15</td><td></td></tr>
</table>
</td></tr>
<tr><td>Test Outcome</td><td>☐Pass | ☐Fail | ☐Conditional Pass | ☐Pass after Fail</td></tr>
</table>

4.6 Laser Plumb2 Brightness

Item Tested:	Laser Plumb2 Brightness PWM Duty cycle
Item Spec:	Laser level cycles duty cycle between 100, 75, 50 when the button is pressed.
Equipment needed:	RC Laser module flashed and programmed, Oscilloscope, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. Laser turns on at desired PWM and outputs a perfect pulse train for detector. **Fail:** 1. Laser doesn’t Turn ON 2. The PWM duty cycle is out of spec.

Test Procedure:

Test Procedure:

- Connect a wire to Test point I133(net label P1). Connect this to oscilloscope to measure freq.
- Set up the oscilloscope to monitor the PWM frequency and duty cycle of the pulse train (200Khz for 500ms and then 8.5Khz for 4ms).
- Connect RC Laser Battery terminal to a Power supply.
- Connect a 10kOhms resistor between B+ and TH on RC Laser Battery terminal.
- Set the power supply to output 20V@300mA

Test Procedure:

1. Insert a charged 20V DeWalt battery pack into the laser unit.
2. Slide the pendulum lock to unlocked position.
3. Press the power button.
4. Turn off any laser that are On.,
5. Press the laser Plumb2 Button.
6. Press the brightness Button.
7. Measure and record the data in result table below.
8. Repeat the process 5 time for each brightness level. i.e Press the Brightness button 15 times.

Output:

S.no	Duty Cycle	Case
1		
2		Test 1
3		
4		
5		Test 2
6		
7		
8		Test 3

9		
10		
11		Test 4
12		
13		
14		Test 5
15		

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

4.7 Last Laser Turn-On

Item Tested:	Last Laser Turn is turned on as soon as the power button is pressed.
Item Spec:	The following takes place after the unit power up: • Level lasers turn ON if all the laser were off when unit was shut down. • Last working laser diode turn on when the unit was shut down.
	RC Laser module flashed and programmed, Oscilloscope, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. The Level laser does turn on by default if all laser diodes were off when shutting down. 2. The last working laser diodes turn on with the Power button. **Fail:** 1. No lasers Turn ON 2. The level laser doesn't turn on if no lasers were on before turning laser unit off 3. The last working lasers do not turn on if there were lasers on before turning laser unit off
Test Procedure:	**Test Procedure:** 1. Connect the dewalt battery to the RC Laser.
Test Procedure:	1. Turn the Unit on by pressing the Power button 2. Level laser should turn on. 3. Press the level laser button to turn the level laser off

4. Repeat from step 1 three times.
5. Record the observation
6. Turn the unit on by pressing the power button
7. Turn additional laser diodes at will
8. Turn the unit off by pressing the power button.
9. Turn the unit On again and record the observation.
10. Repeat from step 6 three times.
11. Repat all the step by replacing turn off sequence to removal of battery.
12. Record the observation.

Output:

S.no	Laser Status	Case
1		Test 1(Power button for default level turn)
2		
3		
4		Test 2(Power button for last laser turned on)
5		
6		
7		Test 3(Battery removal for default laser turn on)
8		
9		
10		Test 4(Battery removal for custom laser turned on)
11		
12		

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

5. Trigger Buttons

5.1 Level

Item Tested:	Level Button – Turn Level Laser ON
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery

<table>
<tr><td>Pass / Fail Criteria</td><td colspan="4">Pass:
1. Level Laser Turned ON when the button is pressed.
2. Level Laser Turns OFF when the button is pressed after the Level Laser is ON.
Fail:
1. Laser doesn’t Turn ON with press of a button
2. Laser doesn’t Turn OFF with press of a button after Level Laser is ON.</td></tr>
<tr><td>Test Procedure:</td><td colspan="4">1. Insert the Dewalt battery pack to RC Laser.
2. Press the Power button
3. Turn Off any laser that are on.
4. Slide the pendulum lock to unlocked position.
5. Press the Laser Level Button to verify function of level laser as ON.
6. Press Button again to verify the level laser if turned OFF.
7. Record the data in result table below.
8. Repeat the above steps and document the observation.
Output:
<table>
<tr><th>Attempt</th><th>Button Action</th><th>State of Level Laser</th></tr>
<tr><td>1</td><td>Button Press (Level Laser OFF state)</td><td></td></tr>
<tr><td></td><td>Button Press</td><td></td></tr>
<tr><td>2</td><td>Button Press (Level Laser OFF state)</td><td></td></tr>
<tr><td></td><td>Button Press</td><td></td></tr>
<tr><td>3</td><td>Button Press (Level Laser OFF state)</td><td></td></tr>
<tr><td></td><td>Button Press</td><td></td></tr>
<tr><td>4</td><td>Button Press (Level Laser OFF state)</td><td></td></tr>
<tr><td></td><td>Button Press</td><td></td></tr>
<tr><td>5</td><td>Button Press (Level Laser OFF state)</td><td></td></tr>
<tr><td></td><td>Button Press</td><td></td></tr>
</table></td></tr>
<tr><td>Test Outcome</td><td>☐Pass</td><td>☐Fail</td><td>☐Conditional Pass</td><td>☐Pass after Fail</td></tr>
</table>

5.2 Plumb1

Item Tested:	Plumb1 Button – Turn Plumb1 Laser ON
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery

Pass / Fail Criteria	**Pass:** 1. Plumb1 Laser Turned ON when the button is pressed. 2. Plumb1 Laser Turns OFF when the button is pressed after the Plumb1 Laser is ON. **Fail:** 1. Laser doesn't Turn ON with press of a button 2. Laser doesn't Turn OFF with press of a button after Plumb1 Laser is ON.
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Press the Power button 3. Turn Off any laser that are on. 4. Slide the pendulum lock to unlocked position. 5. Press the Laser Plumb1 Button to verify function of Plumb1 laser as ON. 6. Press Button again to verify the Plumb1 laser if turned OFF. 7. Record the data in result table below. 8. Repeat the above steps and document the observation. **Output:**

Attempt	**Button Action**	**State of Plumb1 Laser**
1	Button Press (Plumb1 Laser OFF state)	
	Button Press	
2	Button Press (Plumb1 Laser OFF state)	
	Button Press	
3	Button Press (Plumb1 Laser OFF state)	
	Button Press	
4	Button Press (Plumb1 Laser OFF state)	
	Button Press	
5	Button Press (Plumb1 Laser OFF state)	
	Button Press	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

5.3 Plumb2

Item Tested:	Level Button – Turn Plumb2 Laser ON

Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. Plumb2 Laser Turned ON when the button is pressed. 2. Plumb2 Laser Turns OFF when the button is pressed after the Plumb2 Laser is ON. **Fail:** 1. Laser doesn't Turn ON with press of a button 2. Laser doesn't Turn OFF with press of a button after Plumb2 Laser is ON.
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Press the Power button. 3. Turn Off any laser that are on. 4. Slide the pendulum lock to unlocked position. 5. Press the Laser Plumb2 Button to verify function of Plumb2 laser as ON. 6. Press Button again to verify the Plumb2 laser if turned OFF. 7. Record the data in result table below. 8. Repeat the above steps and document the observation. **Output:**

Attempt	Button Action	State of Plumb2 Laser
1	Button Press (Plumb2 Laser OFF state)	
	Button Press	
2	Button Press (Plumb2 Laser OFF state)	
	Button Press	
3	Button Press (Plumb2 Laser OFF state)	
	Button Press	
4	Button Press (Plumb2 Laser OFF state)	
	Button Press	
5	Button Press (Plumb2 Laser OFF state)	
	Button Press	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

5.4 Brightness

Item Tested:	Brightness Button – Change the brightness
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery

Pass / Fail Criteria

Pass:

1. All the lasers cycle the brightness when the button is pressed.

Fail:

1. The Laser remains on one brightness level when the button is pressed.

Test Procedure:

1. Insert the Dewalt battery pack to RC Laser.
2. Press the Power button
3. Turn Off any laser that are on.
4. Slide the pendulum lock to unlocked position.
5. Turn On all the 3 Laser diodes.
6. Press Brightness button to verify the change in brightness.
7. Record the data in result table below.
8. Repeat the above steps and document the observation.

Output:

Attempt	**Button Action**	**Brightness(High,Med ,Low)**
1	Button Press	
	Button Press	
	Button Press	
2	Button Press	
	Button Press	
	Button Press	
3	Button Press	
	Button Press	
	Button Press	
4	Button Press	
	Button Press	
	Button Press	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

5.5 Translate Up

<table>
<tr><td>Item Tested:</td><td>Translate Up Button – Motor Up Movement
Single button press gives out 4 pulses on oscilloscope.
Motor moves up on press and hold on the button.</td></tr>
<tr><td>Equipment needed:</td><td>RC Laser module flashed and programmed, Dewalt Battery</td></tr>
<tr><td>Pass / Fail Criteria</td><td>Pass:
The Unit moves Up on the tracks.
There are 4 pulses sent out on test point after single press.
Fail:
The unit doesn’t move.
There are no pulses sent out on test point after single press.</td></tr>
<tr><td>Test Setup:</td><td>Connect a wire to Test point I55 (net label STEP VERT).
Connect this to oscilloscope to measure freq.
Set up the oscilloscope.
Connect RC Laser terminal to battery pack.</td></tr>
<tr><td>Test Procedure:</td><td>Insert the Dewalt battery pack to RC Laser.
Slide the pendulum lock to unlocked position.
Turn On level laser.
Press the button once and record the number pulses on the oscilloscope.
Record the data in Output table 1 below.
Press and hold Up button to verify its functions.
Record the data in output table 2 below.
Repeat the above steps and document the observation.
Output:<table><tr><th>Attempt</th><th>Button Action</th><th>No. of pulses</th></tr><tr><td>1</td><td>Press button once</td><td></td></tr><tr><td>2</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td></tr><tr><td>5</td><td></td><td></td></tr></table></td></tr>
</table>

<table>
<tr><td></td><td>

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	
2		
3		
4		
5		

Table2

</td></tr>
<tr><td>Test Outcome</td><td>☐Pass ☐Fail ☐Conditional Pass ☐Pass after Fail</td></tr>
</table>

5.6 Translate Down

<table>
<tr><td>Item Tested:</td><td>Translate Down Button – Motor Down Movement
• Single button press gives out 4 pulses on the oscilloscope.
• Motor moves down on press and hold on the button.</td></tr>
<tr><td>Equipment needed:</td><td>RC Laser module flashed and programmed, Dewalt Battery</td></tr>
<tr><td>Pass / Fail Criteria</td><td>Pass:
1. The Unit moves Down on the tracks.
2. There are 4 pulses sent out on test point after single press.
Fail:
1. The unit doesn't move.
1. There are no pulses sent out on test point after single press.</td></tr>
<tr><td>Test setup:</td><td>1. Connect a wire to Test point I55 (net label STEP VERT).
2. Connect this to oscilloscope to measure freq.
3. Set up the oscilloscope.
4. Connect RC Laser terminal to battery pack.</td></tr>
</table>

Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Slide the pendulum lock to unlocked position. 3. Turn On level laser. 4. Press the button once and record the number pulses on the oscilloscope. 5. Record the data in Output table 1 below. 6. Press and hold Up button to verify its functions. 7. Record the data in output table 2 below. 8. Repeat the above steps and document the observation. **Output:** (see Table1 and Table2 below)

Output:

Attempt	Button Action	No. of pulses
1	Press button once	
2		
3		
4		
5		

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	
2		
3		
4		
5		

Table2

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

5.7 Clockwise (>)

Item Tested:	Translate Clockwise Button – Motor Clockwise Movement • Single button press gives out 3 pulses on the oscilloscope. • Motor moves Clockwise on press and hold on the button.
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery

Pass / Fail Criteria	**Pass:** 3. The Unit moves clockwise on the tracks. 4. There are 3 pulses sent out on test point after single press. **Fail:** 2. The unit doesn't move. 1. There are no pulses sent out on test point after single press.
Test setup:	1. Connect a wire to Test point I51 (net label STEP YAW). 2. Connect this to oscilloscope to measure freq. 3. Set up the oscilloscope. 4. Connect RC Laser terminal to battery pack.
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Slide the pendulum lock to unlocked position. 3. Turn On plumb1 laser. 4. Press the button once and record the number pulses on the oscilloscope. 5. Record the data in Output table 1 below. 6. Press and hold Up button to verify its functions. 7. Record the data in output table 2 below. 8. Repeat the above steps and document the observation.

Output:

Attempt	Button Action	No. of pulses
1	Press button once	
2		
3		
4		
5		

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	
2		
3		
4		
5		

	Table2			
Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

5.8 Counterclockwise (<)

Item Tested:	Translate Counterclockwise Button – Motor Counterclockwise Movement • Single button press gives out 3 pulses on the oscilloscope. • Motor moves counterclockwise on press and hold on the button.
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery
Pass / Fail Criteria	**Pass:** 5. The Unit moves counterclockwise on the tracks. 6. There are 3 pulses sent out on test point after single press. **Fail:** 3. The unit doesn't move. 2. There are no pulses sent out on test point after single press.
Test setup:	1. Connect a wire to Test point I50 (net label STEP VERT). 2. Connect this to oscilloscope to measure freq. 3. Set up the oscilloscope. 4. Connect RC Laser terminal to battery pack.
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Slide the pendulum lock to unlocked position. 3. Turn On plumb1 laser. 4. Press the button once and record the number pulses on the oscilloscope. 5. Record the data in Output table 1 below. 6. Press and hold Up button to verify its functions. 7. Record the data in output table 2 below. 8. Repeat the above steps and document the observation. **Output:**

Attempt	Button Action	No. of pulses
1	Press button once	
2		

3		
4		
5		

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	
2		
3		
4		
5		

Table2

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

5.9 Power Button

Item Tested:	Power button – Turns the unit On when the Power button is pressed.
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. The unit Turn ON when Power button is pressed. 2. The unit Turn OFF when the power button is pressed again. **Fail:** 1. The Unit doesn't respond / doesn't Turn ON. 2. The unit doesn't repond/ Turnoff on a consecutive button Press.
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. After the insertion on the pack, The RC module does nothing. 3. Press the power button and the Laser diodes and the Leds on FPC are turned ON. 4. Press the power button again and the Laser Diodes and Leds are turned OFF. 5. Record the data in result table below. 6. Repeat the above steps and document the observation.

	Output:

Attempt	Button Action	State of Unit
1	Button press on	
2	Button press off	
3	Button press on	
4	Button press off	
5	Button press on	
6	Button press off	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

5.10 Pendulum Lock

Item Tested:	Pendulum Lock – Slope led Turn on when pendulum is locked and laser diode have a blink pattern detonating that laser is on slope mode.
Item spec:	The laser blink pattern is as follow: • Laser is solid On for 8.5sec • Laser blinks 3 times for next 1.5sec. This 10sec cycles continues as long pendulum is in locked state.
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. The slope LED turn On when the pendulum lock is in locked position. 2. The laser diodes blink pattern is as explained in the spec. **Fail:** 1. The slope LED does not turn On when the pendulum lock is in locked position. 2. The laser diodes blink pattern is out of spec.
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Slide the pendulum lock to locked position. 3. Press the power button. 4. Turn off any laser that are ON. 5. Turn the level laser on.

	6. Toggle pendulum lock to unlocked position and back to locked position. 7. Record the blink pattern on the level laser. 8. Change the brightness and repeat steps 12 and 13. 9. Record the data in result table below. **Output:**
Test Outcome	☐Pass ☐Fail ☐Conditional Pass ☐Pass after Fail

Attempt	Brightness (High/med/low)	State of blink pattern
1		
2		
3		
4		
5		
6		

6. Status LEDs

6.1 BLE LED

Item Tested:	Battery SOC – LED illumination for Bluetooth status
Equipment needed:	RC Laser module flashed and programmed, Programmable power supply, Dewalt battery pack power supply adapter
Pass / Fail Criteria	**Pass:** 1. The unit displays the correct Bluetooth status **Fail:** 1. ANY time the Bluetooth led does not indicate the correct status.

Test Procedure:	1. Insert the Dewalt battery pack into the laser. 2. Turn the Unit ON by pressing the Power button 3. Observe the status of BLE led. 4. Ble Led blink at 1Hz for 60 sec. 5. Ble LED becomes solid ON when a remote is connected to it. 6. Record the observation. **Output:** **NOTE: LED status options are: ON, OFF, BLINK** **Sn.no** / **BLE Led status**: 1, 2, 3, 4, 5, 6, 7, 8			
Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

Sn.no	BLE Led status
1	
2	
3	
4	
5	
6	
7	
8	

6.2 Battery SOC

Item Tested:	Battery SOC – LED illumination versus battery voltage

Equipment needed:	RC Laser module flashed and programmed, Programmable power supply, Dewalt battery pack power supply adapter
Pass / Fail Criteria	**Pass:** 2. The unit displays the correct SOC LEDs based on battery voltage **Fail:** 2. ANY time the SOC LEDs do not indicate the correct SOC level

Test Procedure:

7. Using the battery pack to power supply adapter, connect the 25V power supply to the RC Laser.
8. Set the power supply current limit to 300mA +/- 50mA.
9. Set the power supply to the first voltage in the table.
10. Record the status of the SOC LEDs in the table.
11. Set the power supply to the second voltage in the table.
12. Record the status of the SOC LEDs in the table.
13. Continue to set the voltages and record the status of the SOC LEDs until all voltages in the table have been tested
14. Repeat the above steps 3 times and document the observation.

Output:

NOTE: LED status options are: ON, OFF, BLINK

Power supply voltage	LED 1 (top)	LED 2 (middle)	LED 3 (bottom)
18.3V +/- 0.05V			
18.1V +/- 0.05V			
16.6V +/- 0.05V			
16.4V +/- 0.05V			
14.9V +/- 0.05V			
14.7V +/- 0.05V			
14.1V +/- 0.05V			
13.9V +/- 0.05V			
13.9V +/- 0.05V			
14.1V +/- 0.05V			
14.7V +/- 0.05V			
14.9V +/- 0.05V			
16.4V +/- 0.05V			
16.6V +/- 0.05V			
18.1V +/- 0.05V			
18.3V +/- 0.05V			

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

6.3 Slope LED

Item Tested:	Slope LED : Slope LED turn on when the pendulum is moved to lock position.
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery
Pass / Fail Criteria	**Pass:** 3. The slope Turn ON when Pendulum is moved locked position. **Fail:** 1. Slope LED doesn't Turn ON.
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Press the power button and turn off any laser diode that are On. 3. Slide the position of pendulum lock switch to locked position. 4. Record the state of Slope LED in result table below. 5. Slide the position of pendulum lock switch o unlocked position. 6. Repeat the above steps and document the observation. **Output:**

Attempt	State of Pendulum	State of LED
1	Locked	
2	Unlocked	
3	Locked	
4	Unlocked	
5	Locked	
6	Unlocked	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

6.4 Drop LED

<table>
<tr><td>Item Tested:</td><td colspan="4">Drop LED – Drop LED turn on when the unit is dropped</td></tr>
<tr><td>Equipment needed:</td><td colspan="4">RC Laser module flashed and programmed, Dewalt Battery</td></tr>
<tr><td>Pass / Fail Criteria</td><td colspan="4">Pass:
1. The unit Drop LED turns ON when Unit is dropped more than 1m.

Fail:
1. The Drop LED doesn’t turn ON when the unit is dropped.</td></tr>
<tr><td>Test Procedure:</td><td colspan="4">1. Insert the Dewalt battery pack to RC Laser.
2. Press the power button again and Turn off any laser that are ON.
3. Drop the Unit on the test bench and record the state of drop led.
4. Repeat the above steps and document the observation.

Output:
<table><tr><th>Attempt</th><th>State of LED</th></tr><tr><td>1</td><td></td></tr><tr><td>2</td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table></td></tr>
<tr><td>Test Outcome</td><td>☐Pass</td><td>☐Fail</td><td>☐Conditional Pass</td><td>☐Pass after Fail</td></tr>
</table>

7. BLE connection

7.1 BLE New remote connection

Item Tested:	BLE LED on Pairing new remote
Equipment needed:	RC Laser and Remote

<table>
<tr><td>Pass / Fail Criteria</td><td>Pass:
1. On pressing Bluetooth button on remote after 60 sec of turning ON the Laser Unit, the BLE connection LED on laser unit is OFF.
2. On pressing Bluetooth button on remote within 60 sec of turning on the laser the BLE connection LED on laser unit STAYS ON.
Fail:
1. On pressing Bluetooth button on remote after 60 sec of turning ON the Laser Unit, the BLE connection LED on laser unit is not OFF.
2. On pressing Bluetooth button on remote within 60 sec of turning on the laser the BLE connection LED on laser unit does not STAY ON.</td></tr>
<tr><td>Test Setup:</td><td>TEST SETUP:
1. Laser unit
2. Remote
3. Timer/Stopwatch</td></tr>
<tr><td>Test Procedure:</td><td>1. Turn on remote by inserting the battery.
2. Turn on the laser and timer for 60 sec.
3. Press and hold the Bluetooth button on remote after 60 sec.
4. Check BLE led.
5. Press and hold the Bluetooth button on remote to unpair the remote.
6. Turn off the laser and then turn on the laser again.
7. Press Bluetooth button on remote before 60 sec.
8. Check BLE led.
9. Repeat the above step and record observation.

Output:
<table>
<tr><th>S.no</th><th>Timer</th><th>State of LED</th></tr>
<tr><td>1</td><td>>60 Sec</td><td></td></tr>
<tr><td></td><td><60sec</td><td></td></tr>
<tr><td></td><td>>60 Sec</td><td></td></tr>
<tr><td></td><td><60sec</td><td></td></tr>
<tr><td></td><td>>60 Sec</td><td></td></tr>
<tr><td></td><td><60sec</td><td></td></tr>
<tr><td></td><td>>60 Sec</td><td></td></tr>
<tr><td></td><td><60sec</td><td></td></tr>
</table></td></tr>
<tr><td>Test Outcome</td><td>☐Pass ☐Fail ☐Conditional Pass ☐Pass after Fail</td></tr>
</table>

7.2 Connection to Paired Remote

Item Tested:	BLE LED turn ON connecting to already paired remote
Equipment needed:	RC Laser and Remote
Pass / Fail Criteria	**Pass:** 1. When laser is turned on with the presence of already paired remote, the blue led (BLE LED) turns ON (solid, not blinking) irrespective of 60 sec time. **Fail:** 1. When laser is turned on with the presence of already paired remote, the blue led (BLE LED) does not turn ON (solid, not blinking) irrespective of 60 sec time.
Test Setup:	**TEST SETUP:** 1. Laser unit 2. Remote 3. Timer/Stopwatch
Test Procedure:	1. Do this after "pair new remote" test 2. Turn on the laser and timer for 60 sec. 3. After 60 sec, turn on remote by inserting the battery. 4. Check BLE led. 5. Turn off the laser and then turn on the laser again (within 60 sec). 6. Check BLE led. 7. Repeat the above step and record observation. **Output:**
Test Outcome	☐Pass ☐Fail ☐Conditional Pass ☐Pass after Fail

Output table (within Test Procedure):

S.no	Timer	State of LED
1	>60 Sec	
	<60sec	
	>60 Sec	
	<60sec	
	>60 Sec	
	<60sec	
	>60 Sec	
	<60sec	

7.3 Unpair a Paired Remote

<table>
<tr><td>Item Tested:</td><td colspan="4">BLE LED turn off when an Already paired remote LRC is Unpaired.</td></tr>
<tr><td>Equipment needed:</td><td colspan="4">RC Laser and Remote</td></tr>
<tr><td>Pass / Fail Criteria</td><td colspan="4">Pass:
1. When an already Paired Remote is Unpaired from the Unit, the BLE LED turn Off indicating a successful unpair.
2. After Unpair the Remote cannot control the Laser
Fail:
1. When an already Paired Remote is Unpaired from the Unit, the BLE LED doesn’t turn Off.
2. After Unpair the Remote can still control the Laser</td></tr>
<tr><td>Test Setup:</td><td colspan="4">TEST SETUP:
1. Laser Unit, Remote, Battery pack, timer.</td></tr>
<tr><td>Test Procedure:</td><td colspan="4">1. Do this after “pair new remote” test
2. Turn on the laser and timer for 60 sec.
3. After 60 sec, turn on remote by inserting the battery.
4. Check BLE led.
5. Press And Hold the BLE Button on the LRC to unpair the remote.
6. Check BLE led.
7. Repair the remote and follow above steps again and note the observation.

Output:
<table><tr><th>Attempt</th><th>State of LED</th></tr><tr><td>1</td><td></td></tr><tr><td>2</td><td></td></tr><tr><td>3</td><td></td></tr><tr><td>4</td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table></td></tr>
<tr><td>Test Outcome</td><td>☐Pass</td><td>☐Fail</td><td>☐Conditional Pass</td><td>☐Pass after Fail</td></tr>
</table>

8. Safety Timeout

8.1 30 sec motor Timeout

Item Tested:	Timeout of vertical and yaw motors
Item Spec:	Motor must stop running after 30 seconds of continuous rotation
Equipment needed:	RC Laser module flashed and programmed, Dewalt battery, Vertical motor, Yaw motor, Timer or stopwatch
Pass / Fail Criteria	**Pass:** 1. The vertical motor stops spinning after 30 seconds +0.5s of continuous rotation 2. The yaw motor stops spinning after 30 seconds +0.5s of continuous rotation **Fail:** 1. The vertical and/or yaw motor continues to spin after 30 seconds of continuous rotation.
Test Setup:	**Test SETUP:** 1. Connect the vertical motor to the vertical motor connector (J5). 2. Connect the yaw motor to the yaw motor connector (J6). 3. Insert the Dewalt battery into the RC Laser unit.
Test Procedure:	**Test PROCEDURE:** 1. Reset the timer. 2. Press and hold the down button on the laser unit. Start the timer as soon as the button is pressed. 3. Record the elapsed time that the vertical motor is spinning. Release the button when the vertical motor stops spinning or after 35 seconds. 4. Reset the timer. 5. Press and hold the up button on the laser unit. Start the timer as soon as the button is pressed. 6. Record the elapsed time that the vertical motor is spinning. Release the button when the vertical motor stops spinning or after 35 seconds. 7. Reset the timer. 8. Press and hold the ‘<’ button on the laser unit. Start the timer as soon as the button is pressed. 9. Record the elapsed time that the yaw motor is spinning. Release the button when the yaw motor stops spinning or after 35 seconds. 10. Reset the timer. 11. Press and hold the ‘>’ button on the laser unit. Start the timer as soon as the button is pressed. 12. Record the elapsed time that the yaw motor is spinning. Release the button when the yaw motor stops spinning or after 35 seconds. **Output:**

Button press	Elapsed time
Down button	
Up button	
‘<’ button	
‘>’ button	

	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

Test Outcome				

8.2 BLE Disconnect Motor Timeout

Item Tested:	BLE Disconnect Stop the any motor that running.
Item Spec:	Motor must stop running as soon as remote disconnects.
Equipment needed:	RC Laser module flashed and programmed, Dewalt battery, Vertical motor, Yaw motor, Timer or stopwatch
Pass / Fail Criteria	**Pass:** 1. The Motor that is running, stop as soon as remote disconnects. **Fail:** 1. The Motor doesn't stop running after the remote disconnects.
Test Setup:	**Test SETUP:** 1. Connect the vertical motor to the vertical motor connector (J5). 2. Connect the yaw motor to the yaw motor connector (J6). 3. Insert the Dewalt battery into the RC Laser unit.

Test Procedure:	**Test PROCEDURE:** 1. Insert the Battery pack in the laser Unit. 2. Press the power button to turn the unit On. 3. Press any button on the LRC so it connects to the Laser unit. 4. Press and hold Up Button to run the vertical motor continuously. 5. Within 30sec, and while holding the up button, remove the Battery from the remote so it disconnects. 6. Observe the state of the motor and record the observation in the table below. 7. Repeat the above with different motor buttons to activate both the motors in all directions. **Output:**

Sno.	Button press	State of motor after remote disconnect
1	Up button	
2		
3	Down Button	
4		
5	‘<’ button	
6		
7	‘>’ button	
8		

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

8.3 Thermistor Cutoff

Item Tested:	Hot pack shutdown – The unit shuts down when the Temp > 70C
Item Spec:	Unit should shut down when the temperature reached > 70C.
Equipment needed:	RC Laser module flashed and programmed, Power supply, Resistance box.
Pass / Fail Criteria	**Pass:** 1. The Turns off, when the temp reaches > 70C(resistance between B+ and TH reaches 2.2k). **Fail:** 1. The unit doesn’t turn off when temp > 70C.

Test Setup:	**Test SETUP:** 1. Set the Power supply to 20V@300mA. 2. Set the resistance box to 10k Ohms. 3. Connect the input of resistance box to B+ and output to TH on Laser unit.
Test Procedure:	**Test PROCEDURE:** 1. Connect B+ and GND on Laser unit to the power supply. 2. Connect the TH terminal on the Laser Unit to the resistance box set at 10k Ohms. 3. Press the power button to turn the unit On. 4. Reduce the Resistance to 2.2k Ohms. 5. Observe the state of the Unit and record the observation in the table below. **Output:**

S.no	Resistance	State of the Unit
1	10	
	2.2	
2	10	
	2.2	
3	10	
	2.2	
4	10	
	2.2	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

8.4 Low Voltage Cutoff

Item Tested:	Low Voltage Cutoff
Item Spec:	Unit should shutdown when the voltage reached below 14.1V
Equipment needed:	RC Laser module flashed and programmed, Power supply,

<table>
<tr><td>Pass / Fail Criteria</td><td>Pass:
1. The Turns off, when the temp reaches < 14.1V

Fail:
1. The unit doesn’t turn off when temp < 14.1V.</td></tr>
<tr><td>Test Setup:</td><td>Test SETUP:

1. Set the Power supply to 20V@300mA.</td></tr>
<tr><td>Test Procedure:</td><td>Test PROCEDURE:

1. Connect B+ and GND on Laser unit to the power supply.
2. Press the Power button to turn the Unit ON.
3. Reduce the voltage down to 14.5V
4. Reduce the voltage down to 13.8V in steps of 0.1V and record the observation when the unit cuts off.
5. Repeat the above step 3 times.

Output:
<table>
<tr><th>S.no</th><th>Volatge</th><th>State of the Unit</th></tr>
<tr><td rowspan="8">1</td><td>14.5</td><td></td></tr>
<tr><td>14.4</td><td></td></tr>
<tr><td>14.3</td><td></td></tr>
<tr><td>14.2</td><td></td></tr>
<tr><td>14.1</td><td></td></tr>
<tr><td>14</td><td></td></tr>
<tr><td>13.9</td><td></td></tr>
<tr><td>13.8</td><td></td></tr>
<tr><td rowspan="8">2</td><td>14.5</td><td></td></tr>
<tr><td>14.4</td><td></td></tr>
<tr><td>14.3</td><td></td></tr>
<tr><td>14.2</td><td></td></tr>
<tr><td>14.1</td><td></td></tr>
<tr><td>14</td><td></td></tr>
<tr><td>13.9</td><td></td></tr>
<tr><td>13.8</td><td></td></tr>
<tr><td rowspan="8">3</td><td>14.5</td><td></td></tr>
<tr><td>14.4</td><td></td></tr>
<tr><td>14.3</td><td></td></tr>
<tr><td>14.2</td><td></td></tr>
<tr><td>14.1</td><td></td></tr>
<tr><td>14</td><td></td></tr>
<tr><td>13.9</td><td></td></tr>
<tr><td>13.8</td><td></td></tr>
</table></td></tr>
<tr><td>Test Outcome</td><td>☐Pass ☐Fail ☐Conditional Pass ☐Pass after Fail</td></tr>
</table>

8.5 Laser Over current Protection Cutoff

<table>
<tr><td>Item Tested:</td><td>Laser Over current protection cutoff</td></tr>
<tr><td>Item Spec:</td><td>Lasers should turnoff when the current reaches above set threshold for each of laser diodes.</td></tr>
<tr><td>Equipment needed:</td><td>RC Laser module flashed and programmed, Power supply,</td></tr>
<tr><td>Pass / Fail Criteria</td><td>Pass:
1. The laser turns off when the current increases beyond the threshold acc to table below.
<table>
<tr><td>Laser diode</td><td>Current cutoff</td></tr>
<tr><td>Level</td><td>350mA</td></tr>
<tr><td>Plumb1</td><td>300mA</td></tr>
<tr><td>Plumb2</td><td>300mA</td></tr>
<tr><td>Level+Plumb1</td><td>650mA</td></tr>
<tr><td>Level+Plumb2</td><td>650mA</td></tr>
<tr><td>Plumb1 + Plumb2</td><td>600mA</td></tr>
<tr><td>Level+plumb1+plumb2</td><td>950mA</td></tr>
<tr><td></td><td></td></tr>
</table>
Fail:
1. The laser do not turn off above a reached threshold.</td></tr>
<tr><td>Test Setup:</td><td>Test SETUP:
1. Set a Power supply at 20V@300mA and connect it to B+ and GND on the terminal.
2. Connect a wire between Test point I136(NET LABEL LASER CURRENT) and another power supply.
3. Connect a wire to Test point I124 (Net label EN_P2)
4. Connect a wire to Test point I128 (Net label EN_L)
5. Connect a wire to Test point I132 (Net label EN_P1)</td></tr>
</table>

Test Procedure:	**Test PROCEDURE:** 6. Press the Power button to turn the Unit ON. 7. Remove the Laser head connector connected on the main PCB board. 8. Disable the power supply at I136. 9. Press the Level Laser button and verify it is on by the signal at test point I128. It should have a PWM signal. 10. Set the power supply connected to I136 to 0.30V and enable the power supply. 11. Increase the voltage on power supply in step of 0.01V until level turns off verified by signal on Test point I128. 12. Disable the power supply connected to I136. 13. Press the Plumb1 Laser button and verify it is on by the signal at test point I132. It should have a PWM signal. 14. Set the power supply connected to I136 to 0.25V and enable the power supply. 15. Increase the voltage on power supply in step of 0.01V until level turns off verified by signal on Test point I136. 16. Disable the power supply connected to I136. 17. Press the Plumb2 Laser button and verify it is on by the signal at test point I124. It should have a PWM signal. 18. Set the power supply connected to I136 to 0.25V and enable the power supply. 19. Increase the voltage on power supply in step of 0.01V until level turns off verified by signal on Test point I124. 20. Disable the power supply connected to I136. 21. Press the Level laser button and the Plumb1 laser button. Verify they are both on by the signals at I128 and I132. They should both have PWM signals. 22. Set the power supply connected to I136 to 0.60V and enable the power supply. 23. Increase the voltage on power supply in step of 0.01V until level and plumb1 turns off verified by signal on Test point I128 and I132. 24. Disable the power supply connected to I136. 25. Press the Level laser and Plumb2 laser buttons. Verify they are both on by the signals at I128 and I132. They should both have PWM signals. 26. Set the power supply connected to I136 to 0.60V and enable the power supply. 27. Increase the voltage on power supply in step of 0.01V until Level and Plumb2 turns off verified by signal on Test point I128 and I132. 28. Disable the power supply connected to I136. 29. Press the Plumb1 laser and Plumb2 laser buttons. Verify they are both on by the signals at I124 and I132. They should both have PWM signals. 30. Set the power supply connected to I136 to 0.55V and enable the power supply. 31. Increase the voltage on power supply in step of 0.01V until Plumb1 and Plumb2 turns off verified by signal on Test point I124 and I132. 32. Disable the power supply connected to I136. 33. Press the Level laser, Plumb1 laser, and Plumb2 laser buttons. Verify they are all on by the signals at I124, I128, and I132. They should all have PWM signals. 34. Set the power supply connected to I136 to 0.90V and enable the power supply. 35. Increase the voltage on step of 0.01V until Level laser, Plumb1 laser, and Plumb2 laser are all off verified by the signals at test points I124, I128, and I132. They should all be <0.2V. 36. Disable the power supply connected to I136. 37. Disable the power supply connected to I92 (Net label B+) 38. Repeat the above step 3 times.

Output:

Trial no	Lasers tested	Voltage at I136	Level laser status (on/off)	Plumb1 laser status (on/off)	Plumb2 laser status (on/off)
1	L	0.30V		N/A	N/A
		0.31V			
		0.32V			
		0.33V			
		0.34V			
		0.35V			
		0.36V			
		0.37V			
		0.38V			
	P1	0.25V	N/A		N/A
		0.26V			
		0.27V			
		0.28V			
		0.29V			
		0.30V			
		0.31V			
		0.32V			
	P2	0.25V	N/A	N/A	
		0.26V			
		0.27V			
		0.28V			
		0.29V			
		0.30V			
		0.31V			
		0.32V			
	L, P1	0.60V			N/A
		0.61V			
		0.62V			
		0.63V			
		0.64V			
		0.65V			
		0.66V			
		0.67V			
	L, P2	0.6V		N/A	
		0.61V			
		0.62V			
		0.63V			
		0.64V			
		0.65V			
		0.66V			
		0.67V			
	P1, P2	0.55V	N/A		
		0.56V			
		0.57V			
		0.58V			

		0.59V			
		0.60V			
		0.61V			
		0.62V			
		0.9V			
		0.91V			
		0.92V			
		0.93V			
		0.94V			
		0.95V			
		0.96V			
	L, P1, P2	0.97V			
2		0.30V		N/A	N/A
		0.31V			
		0.32V			
		0.33V			
		0.34V			
		0.35V			
		0.36V			
		0.37V			
	L	0.38V			
		0.25V	N/A		N/A
		0.26V			
		0.27V			
		0.28V			
		0.29V			
		0.30V			
		0.31V			
	P1	0.32V			
		0.25V	N/A	N/A	
		0.26V			
		0.27V			
		0.28V			
		0.29V			
		0.30V			
		0.31V			
	P2	0.32V			
		0.60V			N/A
		0.61V			
		0.62V			
		0.63V			
		0.64V			
		0.65V			
		0.66V			
	L, P1	0.67V			
		0.6V		N/A	
		0.61V			
	L, P2	0.62V			

			0.63V			
			0.64V			
			0.65V			
			0.66V			
			0.67V			
			0.55V	N/A		
			0.56V			
			0.57V			
			0.58V			
			0.59V			
			0.60V			
			0.61V			
		P1, P2	0.62V			
			0.9V			
			0.91V			
			0.92V			
			0.93V			
			0.94V			
			0.95V			
			0.96V			
		L, P1, P2	0.97V			
	3		0.30V		N/A	N/A
			0.31V			
			0.32V			
			0.33V			
			0.34V			
			0.35V			
			0.36V			
			0.37V			
		L	0.38V			
			0.25V	N/A		N/A
			0.26V			
			0.27V			
			0.28V			
			0.29V			
			0.30V			
			0.31V			
		P1	0.32V			
			0.25V	N/A	N/A	
			0.26V			
			0.27V			
			0.28V			
			0.29V			
			0.30V			
			0.31V			
		P2	0.32V			
			0.60V			N/A
		L, P1	0.61V			

			0.62V			
			0.63V			
			0.64V			
			0.65V			
			0.66V			
			0.67V			
			0.6V		N/A	
			0.61V			
			0.62V			
			0.63V			
			0.64V			
			0.65V			
			0.66V			
		L, P2	0.67V			
			0.55V	N/A		
			0.56V			
			0.57V			
			0.58V			
			0.59V			
			0.60V			
			0.61V			
		P1, P2	0.62V			
			0.9V			
			0.91V			
			0.92V			
			0.93V			
			0.94V			
			0.95V			
			0.96V			
		L, P1, P2	0.97V			

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

9. Accelerometer

9.1 Out of level blink

Item Tested:	Laser diode blink are out of level is detected
Item Spec:	All the laser diodes ON must blink at 1Hz if unit is out of level
Equipment needed:	RC Laser module flashed and programmed, Dewalt battery, Vertical motor, Yaw motor, Timer or stopwatch, remote control

Pass / Fail Criteria	**Pass:** 1. The laser diode blink at 1Hz as soon as Unit is tilted more than 4 degrees. 2. The laser diode blink at 1Hz as soon as the Unit is overturned. **Fail:** 1. The laser diodes do not blink and stay solid on when unit is tilted. 2. The laser diode blink at 1Hz as soon as the Unit is overturned.
Test Setup:	**Test SETUP:** 1. Connect the Laser head cable(white cable) in the J7 connector on the main unit board. 2. Connect the Dewalt battery. 3. Remote connection is not necessary for this test.
Test Procedure:	**Test PROCEDURE:** 1. Turn the unit ON by pressing the Power On button 2. By default Laser Level should turn ON. 3. Turn ON the plumb1 and plumb 2 laser ON. 4. Tilt the unit to > 4 degree with battery as a pivot point on the xaxis. 5. Record the observation 6. Return the unit back to 0 degree and record the observation 7. Tilt the unit to > 4 degree with battery as a pivot point on the yaxis. 8. Record the Observation. 9. Overturn the unit where unit becomes Upside down 10. Record the observation 11. Return the unit back to 0 degree where the unit is resting on the battery. 12. Repeat the above steps 3 times **Output:**

X axis	Blink(yes/no)	Y axis	Blink(yes/no)	Z axis	Blink(yes/no)
4		4		Overturned	
0		0		0	
4		4		Overturned	
0		0		0	
4		4		Overturned	
0		0		0	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

10. Laser control through Remote

10.1 Level

Item Tested:	Level Laser operation by remote

Item Spec:	Level Laser turns ON and OFF using the remote control			
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote			
Pass / Fail Criteria	**Pass:** 2. Level Laser turn ON or OFF using the remote control **Fail:** 3. Level laser doesn't turn ON when it is off and the Level Laser button is pressed. 4. Level laser doesn't turn OFF when it is on and the Level Laser button is pressed.			
Test Procedure:	1. Insert the 20V DeWalt battery pack into the laser unit. 2. Slide the pendulum lock to unlocked position. 3. Press the power button on the laser unit. 4. Turn off any lasers that are on. 5. Press the Level Laser Button on the remote. 6. Record the data in the result table below. 7. Press the Level Laser Button on the remote. 8. Record the data in the result table below. 9. Repeat steps 5-8 two more times (total 6 button presses). **Output:** Press number / Level laser status (ON/OFF): 1 / ; 2 / ; 3 / ; 4 / ; 5 / ; 6 /			
Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

Output:

Press number	**Level laser status (ON/OFF)**
1	
2	
3	
4	
5	
6	

10.2 Plumb1

Item Tested:	Plumb1 Laser operation by remote
Item Spec:	Plumb1 Laser turns ON and OFF using the remote control
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote

Pass / Fail Criteria	**Pass:** 3. Plumb1 Laser turn ON or OFF using the remote control **Fail:** 5. Plumb1 laser doesn't turn ON when it is off and the Plumb1 Laser button is pressed. 6. Plumb1 laser doesn't turn OFF when it is on and the Plumb1 Laser button is pressed.
Test Procedure:	10. Insert the 20V DeWalt battery pack into the laser unit. 11. Slide the pendulum lock to unlocked position. 12. Press the power button on the laser unit. 13. Turn off any lasers that are on. 14. Press the Plumb1 Laser Button on the remote. 15. Record the data in the result table below. 16. Press the Plumb1 Laser Button on the remote. 17. Record the data in the result table below. 18. Repeat steps 5-8 two more times (total 6 button presses). **Output:**
Test Outcome	☐Pass ☐Fail ☐Conditional Pass ☐Pass after Fail

Press number	Plumb1 Laser status (ON/OFF)
1	
2	
3	
4	
5	
6	

10.3 Plumb2

Item Tested:	Plumb2 Laser operation by remote
Item Spec:	Plumb2 Laser turns ON and OFF using the remote control
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote

Pass / Fail Criteria	**Pass:** 4. Plumb2 Laser turn ON or OFF using the remote control **Fail:** 7. Plumb2 laser doesn't turn ON when it is off and the Plumb2 Laser button is pressed. 8. Plumb2 laser doesn't turn OFF when it is on and the Plumb2 Laser button is pressed.
Test Procedure:	19. Insert the 20V DeWalt battery pack into the laser unit. 20. Slide the pendulum lock to unlocked position. 21. Press the power button on the laser unit. 22. Turn off any lasers that are on. 23. Press the Plumb2 Laser Button on the remote. 24. Record the data in the result table below. 25. Press the Plumb2 Laser Button on the remote. 26. Record the data in the result table below. 27. Repeat steps 5-8 two more times (total 6 button presses). **Output:**

Press number	Level laser status (ON/OFF)
1	
2	
3	
4	
5	
6	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

10.4 Translate Up

Item Tested:	Translate Up operation by remote • Single button press gives out 4 pulses on oscilloscope. • Motor moves up on press and hold on the button.
Item Spec:	Unit moves UP using the remote control
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote

Pass / Fail Criteria

Pass:

1. The Unit moves Up on the tracks.
2. There are 4 pulses sent out on test point after single press.

Fail:

1. The unit doesn’t move.
1. There are no pulses sent out on test point after single press.

Test Setup:

1. Connect a wire to Test point I55 (net label STEP VERT).
2. Connect this to oscilloscope to measure freq.
3. Set up the oscilloscope.
4. Connect RC Laser terminal to battery pack.

Test Procedure:

1. Insert the Dewalt battery pack to RC Laser.
2. Slide the pendulum lock to unlocked position.
3. Turn On level laser.
4. Press the button once and record the number pulses on the oscilloscope.
5. Record the data in Output table 1 below.
6. Press and hold Up button to verify its functions.
7. Record the data in output table 2 below.
8. Repeat the above steps and document the observation.

Output:

Attempt	Button Action	No. of pulses
1	Press button once	
2		
3		
4		
5		

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	
2		
3		
4		
5		

	Table2			
Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

10.5 Translate Down

Item Tested:	Translate Down operation by remote • Single button press gives out 4 pulses on oscilloscope. • Motor moves up on press and hold on the button.
Item Spec:	Unit moves down using the remote control
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery, remote
Pass / Fail Criteria	**Pass:** 1. The Unit moves Down on the tracks. 2. There are 4 pulses sent out on test point after single press. **Fail:** 1. The unit doesn't move. 2. There are no pulses sent out on test point after single press.
Test Setup:	1. Connect a wire to Test point I55 (net label STEP VERT). 2. Connect this to oscilloscope to measure freq. 3. Set up the oscilloscope. 4. Connect RC Laser terminal to battery pack.
Test Procedure:	1. Insert the Dewalt battery pack to RC Laser. 2. Slide the pendulum lock to unlocked position. 3. Turn On level laser. 4. Press the button once and record the number pulses on the oscilloscope. 5. Record the data in Output table 1 below. 6. Press and hold Up button to verify its functions. 7. Record the data in output table 2 below. 8. Repeat the above steps and document the observation.

Output:

Attempt	Button Action	No. of pulses
1	Press button once	
2		
3		
4		
5		

Table1

Attempt	Button Action	State of Unit
1	Button press and hold	
2		
3		
4		
5		

Table2

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

10.6 Clockwise(>)

Item Tested:	Translate Clockwise Opertaionby Remote • Single button press gives out 3 pulses on the oscilloscope. • Motor moves clockwise on press and hold on the button.
Equipment needed:	RC Laser module flashed and programmed, Dewalt Battery
Pass / Fail Criteria	**Pass:** 1. The Unit moves clockwise on the tracks. 2. There are 3 pulses sent out on test point after single press. **Fail:** 1. The unit doesn't move. 3. There are no pulses sent out on test point after single press.

Test setup:

1. Connect a wire to Test point I50 (net label STEP VERT).
2. Connect this to oscilloscope to measure freq.
3. Set up the oscilloscope.
4. Connect RC Laser terminal to battery pack.

Test Procedure:

1. Insert the Dewalt battery pack to RC Laser.
2. Slide the pendulum lock to unlocked position.
3. Turn On plumb1 laser.
4. Press the button once and record the number pulses on the oscilloscope.
5. Record the data in Output table 1 below.
6. Press and hold Up button to verify its functions.
7. Record the data in output table 2 below.
8. Repeat the above steps and document the observation.

Output:

Attempt	**Button Action**	**No. of pulses**
1	Press button once	
2		
3		
4		
5		

Table1

Attempt	**Button Action**	**State of Unit**
1	Button press and hold	
2		
3		
4		
5		

Table2

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

10.7 Counterclockwise (<)

<table>
<tr><td>Item Tested:</td><td>Translate Counter Clockwise operation by remote
Single button press gives out 3 pulses on the oscilloscope.
Motor moves lockwise on press and hold on the button.</td></tr>
<tr><td>Equipment needed:</td><td>RC Laser module flashed and programmed, Dewalt Battery</td></tr>
<tr><td>Pass / Fail Criteria</td><td>Pass:
1. The Unit moves clockwise on the tracks.
2. There are 3 pulses sent out on test point after single press.
Fail:
1. The unit doesn’t move.
4. There are no pulses sent out on test point after single press.</td></tr>
<tr><td>Test setup:</td><td>1. Connect a wire to Test point I50 (net label STEP VERT).
2. Connect this to oscilloscope to measure freq.
3. Set up the oscilloscope.
4. Connect RC Laser terminal to battery pack.</td></tr>
<tr><td>Test Procedure:</td><td>1. Insert the Dewalt battery pack to RC Laser.
2. Slide the pendulum lock to unlocked position.
3. Turn On plumb1 laser.
4. Press the button once and record the number pulses on the oscilloscope.
5. Record the data in Output table 1 below.
6. Press and hold Up button to verify its functions.
7. Record the data in output table 2 below.
8. Repeat the above steps and document the observation.

Output:
<table>
<tr><th>Attempt</th><th>Button Action</th><th>No. of pulses</th></tr>
<tr><td>1</td><td>Press button once</td><td></td></tr>
<tr><td>2</td><td></td><td></td></tr>
<tr><td>3</td><td></td><td></td></tr>
<tr><td>4</td><td></td><td></td></tr>
<tr><td>5</td><td></td><td></td></tr>
</table>
Table1
<table>
<tr><th>Attempt</th><th>Button Action</th><th>State of Unit</th></tr>
<tr><td>1</td><td>Button press and hold</td><td></td></tr>
<tr><td>2</td><td></td><td></td></tr>
<tr><td>3</td><td></td><td></td></tr>
<tr><td>4</td><td></td><td></td></tr>
</table>
</td></tr>
</table>

5		

Table2

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

11. Datalogging

Item Tested:	Datalogging parameters.
Item Spec:	Datalogging works with the SBD Ibox and the parameters are changing per runtime.
Equipment needed:	RC Laser module flashed and programmed, Dewalt batter, SBD IBox.
Pass / Fail Criteria	**Pass:** 1. Parameters change per run using when read through Ibox. **Fail:** 1. Parameter do not change per run when read through IBox.

Test Procedure:	**Test PROCEDURE:** 1. Insert the DeWalt pack to Ibox. 2. Connect the IBox to the computer and start the bootloader app. 3. Insert the Ibox in the laser unit. 4. Press the power button tot turn the unit ON. 5. Read the Datalog information from the bootloader app. 6. Note the value of the switch cycle and runtime. 7. Remove the ibox from the unit , insert the pack and press the Laser level , plumb1 button. 8. Follow steps from 3 to 6 and record the observation in the output section. 9. Share the snippet from the bootloader section. **Output:**			
Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail

12. Calibration

Item Tested:	Calibration using the IBox.
Equipment needed:	RC Laser module flashed and programmed, Dewalt battery, SBD IBox.
Pass / Fail Criteria	**Pass:** 1. Calibration complete **Fail:** 1. Calibration incomplete

Test Procedure:	**Test PROCEDURE:** 1. Insert the DeWalt pack to Ibox. 2. Connect the IBox to the computer and start the bootloader app. 3. Insert the Ibox in the laser unit. 4. Press the power button to turn the unit ON. 5. In the Bootloader app, press the Calibrate button as shown in the image above. 6. Notice the state of the unit. 7. Level laser turn on indicating the start of calibration process. 8. After some time, plumb/vertical laser turn on Indicating the end of Calibration. 9. After some time, all the laser turn off indicating the calibration routine has been completed. 10. Repeat the above steps 3 times and record in table below **Output:**

Sno	Unit Calibrated
1	
2	
3	

Test Outcome	☐Pass	☐Fail	☐Conditional Pass	☐Pass after Fail